Partially Identified: privacy suppression and the automation–augmentation measure in the Anthropic Economic Index

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Abstract

The Anthropic Economic Index (AEI) reports, for each O*NET task, the share of observed AI usage that is automation rather than augmentation. The measure has been adopted quickly: it underpins occupation-level wage regressions, an NBER working paper on task chaining, and regional exposure indices marketed for workforce-transition targeting. This paper shows that the measure is partially identified. AEI suppresses any cell holding fewer than 15 conversations from fewer than five accounts, which censors the data on two margins, i.e. modes within observed tasks and whole tasks within occupations. Bounding both, 42 of 775 six-digit SOC occupations — 5.4% — have an identified set narrower than 0.10 on at least five observed tasks, and sixteen of 22 major groups contain none. The identified set is narrow only for text-mediated cognitive work and effectively spans the unit interval for farming, construction, transportation, and installation and repair. Suppression removes just 6.38% of classified conversation volume; the loss of identification is far larger because suppression concentrates in thin cells and most tasks are thin. Identification is a property of data density rather than of the work: log classified volume alone explains 76% of the variation in joint bound width, and 21 occupational dummies add 1.2 percentage points on top of volume and coverage.

Keywords: partial identification, statistical disclosure control, AI exposure, measurement error, O*NET
JEL: C18, C81, J24, O33

1 Introduction: a measure of behaviour, not capability

Measures of AI exposure have multiplied, and nearly all of them are measures of *capability*: what a model could in principle perform, imputed from O*NET descriptions by expert judgment or by model scoring. Frey and Osborne (2017), Brynjolfsson, Mitchell and Rock (2018), Webb (2020), Felten, Raj and Seamans (2021) and Eloundou et al. (2024) all share that structure, and they share its central weakness, i.e. that the object being measured is a counterfactual about what could happen rather than an observation of what does. The Anthropic Economic Index is different in kind. It reports observed behaviour — what users actually delegate to a model — and, uniquely among the public measures, classifies each conversation by interaction mode, which permits a split between *automation*, meaning full delegation, and *augmentation*, meaning collaboration. That split maps directly onto the theoretical distinction between displacement and reinstatement, which is what makes it valuable and what explains the speed of its uptake.

The uptake is real and growing. Chanoi and Bangert-Drowns (2025) relate occupational wages to AI exposure constructed from the index. Demirer et al. (2026) use the execution labels to identify chains of tasks that can be handed to a model as a block. Brynjolfsson, Chandar and Chen (2025) use the substitution–complementarity distinction to interpret differential employment movements among young workers. At least two regional indices published in 2026 employment-weight AEI scores to metropolitan and county level, explicitly for workforce-transition targeting.

This paper argues that the split cannot bear that weight, and it is worth being precise about why, because the reason is neither that the underlying classification is wrong nor that the privacy protection is inappropriate. The protection is appropriate, and Anthropic documents it clearly. The problem is that the released file has passed through a disclosure-control filter that removes any cell holding fewer than 15 conversations from fewer than five accounts, and that downstream users read the resulting absences as measured zeros rather than as missing data. We treat suppression as interval censoring and derive worst-case bounds on the automation share at both task and occupation level. The parameter that most users treat as a point estimate is, for the large majority of occupations, known only to lie in an interval covering much of its range.

Three findings follow. **First**, the censoring is mechanical rather than incidental: the number of collaboration categories reported for a task is almost perfectly determined by that task's conversation volume, the minimum volume in each stratum rising 15, 17, 33, 49, 69 and 141 across strata holding one through six categories. Mode coverage therefore carries no information about a task's actual interaction profile. **Second**, the resulting bounds are wide; median width is 0.5411 at task level and aggregation to six-digit SOC narrows it only to 0.4177, while just 238 of 2,616 observed tasks — 1.3% of the O*NET universe — are point-identified. **Third**, the loss of identification is systematically distributed, and in the direction least convenient for policy. Median width is 0.064 for Computer and Mathematical occupations and 0.089 for Educational Instruction, against 0.73 to 1.00 for Farming, Construction, Installation and Repair, Protective Service and Personal Care. The automation-versus-augmentation distinction is identified precisely where it was least contested, and unidentified where the policy question is live.

Adding the extensive margin sharpens all three. Once tasks that never appear in the released file are censored alongside modes that were dropped from tasks that do, 225 of 775 SOC codes have no computable share at all, the median occupation could be concealing 59.9% of its task volume, and only 42 occupations — 5.4% — retain an identified set narrower than 0.10 on at least five observed tasks.

A methodological note may be the paper's most useful practical contribution. Two errors in a naive reconstruction very nearly cancel: suppression biases a task-level reconstruction of the global automation share upward by 0.550 percentage points, while failing to exclude the `none` pseudo-task biases it downward by 0.472. A researcher who commits the second error while unaware of the first recovers Anthropic's published figure to within 0.078 percentage points and concludes that the pipeline validates. It does not. That near-cancellation is, we suspect, part of why the problem has gone unremarked.

2 Related work: selection, disclosure, and partial identification

Three literatures meet here, and the contribution sits in the gap between them.

Critiques of exposure measures. The Budget Lab at Yale (2026) harmonises seven AI exposure metrics and finds that they agree about which occupations are *not* exposed while diverging sharply

about the highly exposed, with variance rising in exposure. That gradient runs opposite to ours. Their metrics disagree most where our bounds are tightest, because text-mediated cognitive work is both contested across measures and densely observed in usage data; their metrics agree where our bounds are useless, because manual work is uncontroversially unexposed on capability grounds and almost entirely absent from usage data. Neither end of the distribution is well measured, for opposite reasons, which makes this paper a complement to theirs rather than a competitor.

Selection in usage data. Yin and Ogut (2026) formalise platform selection as non-classical measurement error and construct workforce-reweighted partial-identification bounds by reweighting platform shares to BLS employment shares, reporting attenuation of 42 to 93 percent. Theirs is the closest methodological cousin, and the distinction is worth stating plainly: they bound the consequences of *who is on the platform*, whereas we bound the consequences of *what the disclosure filter removed from the released file*. The two mechanisms are orthogonal and both apply, so the bounds compose rather than compete.

Rater dependence in judged measures. Rubric-based exposure scores, in which a model rates each O*NET task directly, are the obvious alternative to observed usage, and they carry an identification problem of their own. Yin, Vu and Persico (2026) apply the Eloundou et al. (2024) rubric with three frontier models to identical task data and find mean exposure varying 3.6-fold across them. Selection into the platform, the choice of rater, and disclosure censoring are thus three distinct sources of error in AI exposure measurement; this paper quantifies the third, and Section 7.5 returns to the second.

Users of the split. Demirer et al. (2026) provide the clearest illustration of the problem and, to their credit, the clearest documentation of it. They report that of 3,364 tasks linked to conversations, 1,017 have all conversations filtered, and they then state that they "treat tasks that do not appear in the Anthropic dataset, as well as those with 100% filtered conversations, as manual tasks," describing this as a conservative measure that limits statistical power. The choice is defensible and openly declared. The point we wish to make is narrower and, we think, fair: recode-to-manual is not a neutral default but an identifying assumption, and it selects one endpoint of the identified set we derive below.

Statistical disclosure control. Cox (1980, 1995) established cell suppression as the standard protection for tabular releases, and Fischetti and Salazar-González (2003) proposed partial suppression, which replaces suppressed entries with intervals containing the true value. That is essentially the move we make here, though from the outside: the agency published suppressed cells, and we reconstruct the intervals the agency would have published had it adopted the partial-suppression convention.

Partial identification. The bounds themselves are elementary applications of the framework in Manski (2003), with the worst-case logic of Horowitz and Manski (1995) on contaminated and corrupted data and the interval-data treatment of Manski and Tamer (2002). Where sampling variability enters, at the aggregation step, Imbens and Manski (2004) and Stoye (2009) supply the appropriate confidence intervals.

Table 1: Collaboration categories in the `onet_task::collaboration` intersection, by number of tasks carrying the category. Coverage is out of 2,616 observed tasks.

Category	Tasks	Coverage
`none`	222	0.085
feedback loop	367	0.140
validation	412	0.157
learning	1,033	0.395
task iteration	1,127	0.431
directive	1,584	0.606
`not_classified`	2,476	0.946

3 Data and the suppression mechanism: seven categories, not five

3.1 The release

We use AEI `release_2025_09_15`, the enriched Claude.ai file, global aggregate, sampling 4–11 August 2025, licensed CC-BY 4.0. The file contains 136,845 rows, of which 29,681 are the global subset. Occupation linkage uses `onet_task_statements.csv` from `release_2025_02_10`, which supplies 19,530 occupation–task rows over 18,428 distinct normalised task statements.[1]

Two absences in the facet inventory matter for interpretation and are worth recording. There is no usage-context facet separating work from personal or coursework usage, and there is no `request::onet_task` intersection, so non-work usage cannot be filtered out of the task measure in this release.[2]

3.2 The collaboration dimension has seven values

Anthropic documents five interaction modes: directive, feedback loop, task iteration, validation and learning. The released intersection file carries seven. Alongside the five documented modes it contains `not_classified`, which appears for 94.6% of tasks, and `none`, which appears for 8.5%. Table 1 gives the full distribution.

The mean number of *classified* modes present per task is 1.729 of five. Neither `none` nor `not_classified` is a mode: the first records conversations the classifier judged to match none of the five, the second records conversations withheld or unclassifiable. Both must be excluded from any automation share, and we exclude them throughout.[3]

[1] The occupational universe throughout is the 775 distinct six-digit SOC codes obtained from the 974 eight-digit O*NET-SOC codes appearing in that file, after stripping the `.00` and `.01` extensions O*NET uses for sub-occupations. This is narrower than the full 2019 O*NET-SOC detailed taxonomy, and we take the AEI file's own universe as the relevant denominator, since an occupation absent from that file could not appear in the released usage data under any circumstances. Using O*NET's full list would lower every coverage percentage reported below rather than raise it.

[2] This is a limitation of the release rather than of the index. It bears on level comparisons between AEI shares and employment-weighted quantities, but not on the identification argument, which concerns which cells survive the filter rather than which conversations entered them.

[3] Removing them costs six intersection rows and leaves the observed task universe unchanged at 2,616. The consequence of *not* removing `none` is quantified in Section 6, and it is substantial.

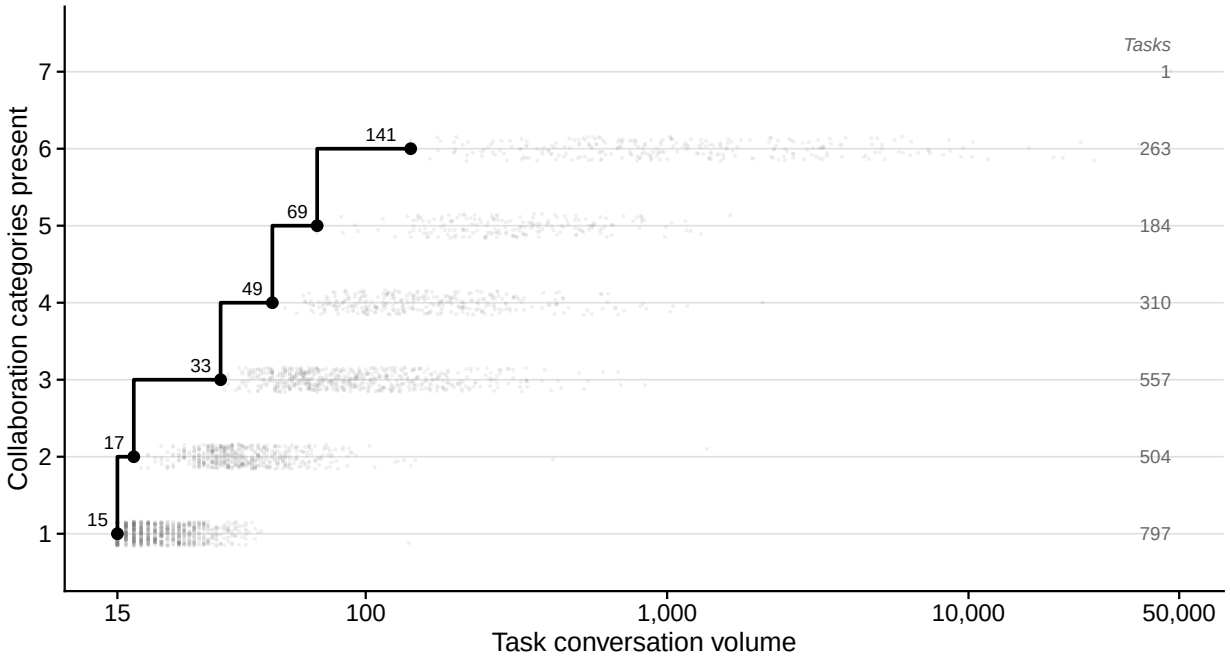


Figure 1: Collaboration-mode coverage against task conversation volume

Notes: Each grey point is one of the 2,616 O*NET tasks with observed usage, positioned by its total conversation volume (log scale) and the number of collaboration categories reported for it; points are jittered vertically for legibility. The stepped line traces the minimum volume observed in each stratum: 15, 17, 33, 49, 69 and 141 conversations for one through six categories. Right-hand figures give the number of tasks in each stratum. The monotone left edge is the signature of a minimum-cell-size disclosure rule rather than a property of the tasks: a task must be larger before more of its modes clear the threshold and survive into the released data. The single seven-category task is the highest-volume task in the release.
Source: Anthropic Economic Index, release 2025-09-15 (CC-BY 4.0); author's calculations.

3.3 The privacy floor

Anthropic filters cells holding fewer than 15 conversations from fewer than five unique accounts. Applied to the collaboration dimension, this removes 59,135 of 927,255 classified conversations, i.e. 6.377% of classified volume, leaving 868,120 conversations inside surviving cells.

Figure 1 plots the number of categories present against task volume on a log scale, and the strata are clean. Their lower bounds step up 15, 17, 33, 49, 69 and 141 for one through six categories, which is the arithmetic signature of a minimum-cell-size rule rather than of any property of the underlying interaction profile. A task is not reported as single-mode because its users interact with the model in one way; it is reported as single-mode because it is small.

3.4 Identifying assumption

The filter has two criteria and we treat the conversation criterion as binding, so that a suppressed cell holds at most 14 conversations. If the account criterion binds for some cells, i.e. cells with relatively high volume concentrated in few accounts, then the true hidden mass exceeds 14 and our bounds are too narrow. The bounds are therefore conservative in the direction that matters. Every statement below should be read as a lower bound on the severity of the problem rather than as an

Table 2: Bound width by number of surviving classified modes. Tasks with $k = 0$ are width 1 by construction; the algebraic maximum at $k = 1$ is $56/71 = 0.7887$.

k	Tasks	Median width	Max width
0	809	1.000	1.000
1	506	0.737	0.789
2	556	0.382	0.583
3	313	0.174	0.368
4	194	0.044	0.173
5	238	0.000	0.000

estimate of it.

4 The intensive margin: modes censored within observed tasks

4.1 Construction

Let task i report k_i of the five classified modes, with a_i conversations in the automation modes (directive, feedback loop) and g_i in the augmentation modes (validation, task iteration, learning). Each of the $5 - k_i$ absent modes holds between 0 and C conversations, with $C = 14$. Write the censored mass as $c_i = (5 - k_i)C$. Assigning all of it to automation gives the upper bound and all of it to augmentation the lower:

$$\underline{\theta}_i \;=\; \frac{a_i}{a_i + g_i + c_i}, \qquad \overline{\theta}_i \;=\; \frac{a_i + c_i}{a_i + g_i + c_i}, \qquad \overline{\theta}_i - \underline{\theta}_i \;=\; \frac{c_i}{a_i + g_i + c_i}.$$

This is an identified set, not a confidence interval. Both endpoints are known given the threshold, so the uncertainty here is deterministic rather than sampling variability, and no distributional assumption is involved.[4]

4.2 Results

Across 2,616 observed tasks the median bound width is 0.5411 and the mean 0.5481, with an interquartile range of 0.1793 to 1.000. Table 2 gives the ladder in k.

The ladder is perfectly monotone, and the maximum at $k = 1$ attains the algebraic ceiling of 56/71, which is a useful internal check that the arithmetic is doing what it should. Two counts deserve emphasis. First, 809 tasks report no classified mode at all and are wholly uninformative about the split. Second, only 238 tasks are point-identified, which is 9.1% of observed tasks and 1.3% of the O*NET universe.

A further 516 of the 1,807 tasks reporting at least one classified mode are degenerate, in the sense that all surviving volume falls on one side of the split and the observed share is exactly 0 or 1. Removing these leaves 1,291 tasks, i.e. 49.4% of observed tasks and 7.0% of the O*NET universe, on which a non-degenerate share can be computed at all.

[4]Sampling variation does enter at the aggregation step of Section 4.4, where task-level quantities are combined across occupations with an equal-split rule. A referee at an econometrics venue will reasonably expect Imbens–Manski or Stoye intervals there, and we regard adding them as the natural next step rather than as an objection to the construction.

4.3 Sensitivity to the censoring cap

The bounds depend on C, and it is fair to ask whether the widths reported above are an artefact of setting it at 14. However, the sweep says otherwise. Median width is 0.2963 at $C = 5$, 0.4572 at $C = 10$, 0.5411 at $C = 14$ and 0.5565 at $C = 14.9$. Even under the implausibly generous assumption that suppressed cells hold at most five conversations, the median identified set still spans nearly thirty points of the share.

4.4 Aggregation to occupations

Following the equal-split convention, a task's volume divides equally across every occupation whose O*NET profile lists it. Writing n_t for the number of occupations listing task t and T_j for the tasks listed by occupation j,

$$\underline{\theta}_j = \frac{\sum_{t \in T_j} a_t / n_t}{\sum_{t \in T_j} (a_t + g_t + c_t)/n_t}, \qquad \overline{\theta}_j = \frac{\sum_{t \in T_j} (a_t + c_t)/n_t}{\sum_{t \in T_j} (a_t + g_t + c_t)/n_t}.$$

The task-to-O*NET join matches 2,616 of 2,616 observed tasks, a rate of 100%, so no attrition enters here. Of 775 six-digit SOC codes, 550 have at least one task carrying mode data. Among those, median width is 0.4177 and mean 0.4703. Aggregation narrows the bounds by roughly a fifth relative to the task level, which is real but modest; averaging over tasks cannot recover a parameter that was never observed in the first place, it can only spread the ignorance more evenly.

4.5 Major-group medians conceal large within-group spread

Table 3 orders the 22 major groups by median intensive-margin width, and the ordering is the paper's central substantive result in miniature.

The ordering is not occupational in any meaningful sense. It is a ranking of how text-mediated the work is, i.e. of how much of it passes through a chat interface at all, and the apparent occupational structure is a consequence of that rather than a cause of it. That reading is formalised in Section 5.4.

5 The extensive margin: tasks censored within occupations

5.1 The second channel

Section 4 censors modes within tasks that appear in the released file; however, it says nothing at all about tasks that never appear, and those turn out to be the larger problem. An occupation whose O*NET profile lists forty tasks, of which three survive into the release, has an intensive-margin bound computed from three tasks and thirty-seven silences. Treating the thirty-seven as zero volume is the same error as treating a suppressed mode as zero volume, one level up.

5.2 Construction

Each unobserved task in an occupation's profile holds between 0 and C conversations on the same reasoning as before. Writing v_j^{cls} for observed classified volume, c_j^{int} for intensive censored mass

Table 3: Intensive-margin bounds by SOC major group, ordered by median width. Restricted to the 550 codes with at least one observed task carrying mode data.

Major	Group	Occs	Median width	Median lo	Median hi
15	Computer and Mathematical	18	0.064	0.468	0.607
25	Educational Instruction and Library	60	0.089	0.283	0.405
37	Building and Grounds Maintenance	5	0.215	0.318	0.611
19	Life, Physical, and Social Science	42	0.216	0.281	0.574
23	Legal	8	0.218	0.396	0.720
41	Sales and Related	19	0.321	0.316	0.653
27	Arts, Design, Entertainment, Media	31	0.358	0.323	0.680
35	Food Preparation and Serving	13	0.396	0.306	0.741
21	Community and Social Service	13	0.417	0.209	0.647
43	Office and Administrative Support	50	0.431	0.298	0.789
13	Business and Financial Operations	29	0.452	0.257	0.701
11	Management	32	0.467	0.251	0.736
17	Architecture and Engineering	29	0.498	0.202	0.712
29	Healthcare Practitioners and Technical	47	0.512	0.157	0.676
51	Production	36	0.692	0.192	0.880
31	Healthcare Support	14	0.718	0.138	0.774
53	Transportation and Material Moving	20	0.721	0.262	1.000
39	Personal Care and Service	24	0.732	0.181	0.872
33	Protective Service	13	0.767	0.162	1.000
47	Construction and Extraction	21	0.778	0.129	1.000
49	Installation, Maintenance, and Repair	23	0.778	0.000	0.882
45	Farming, Fishing, and Forestry	3	1.000	0.000	1.000

and c_j^{ext} for extensive censored mass, all equal-split, the joint bounds are

$$\underline{\theta}_j^{\text{joint}} = \frac{a_j}{v_j^{\text{cls}} + c_j^{\text{int}} + c_j^{\text{ext}}}, \qquad \overline{\theta}_j^{\text{joint}} = \frac{a_j + c_j^{\text{int}} + c_j^{\text{ext}}}{v_j^{\text{cls}} + c_j^{\text{int}} + c_j^{\text{ext}}}.$$

We also report $h_j = c_j^{\text{ext}} / (v_j^{\text{obs}} + c_j^{\text{ext}})$, the maximum share of an occupation's task volume that suppression could be concealing.

5.3 Results

Of 775 SOC codes, 225 — 29.0% — have no task carrying any mode data, so no point estimate of the automation share can be formed for them at all. The joint bound is nonetheless defined, and it is worth stating what it equals: with no observed automation volume and no observed classified volume, the identified set for each of these 225 occupations is exactly $[0, 1]$. Every value of the automation share is consistent with the released data. This is the sense in which the extensive margin is not a matter of degree, i.e. for close to three occupations in ten the measure is altogether silent. For the median occupation, $h_j = 0.599$: suppression could be concealing three-fifths of its task volume. Some 57.3% of occupations exceed one half.

Joint bound widths are correspondingly wide. Across all 775 codes the median is 0.883; among the 550 with any data it is 0.682. Adding the extensive margin to the intensive one widens the

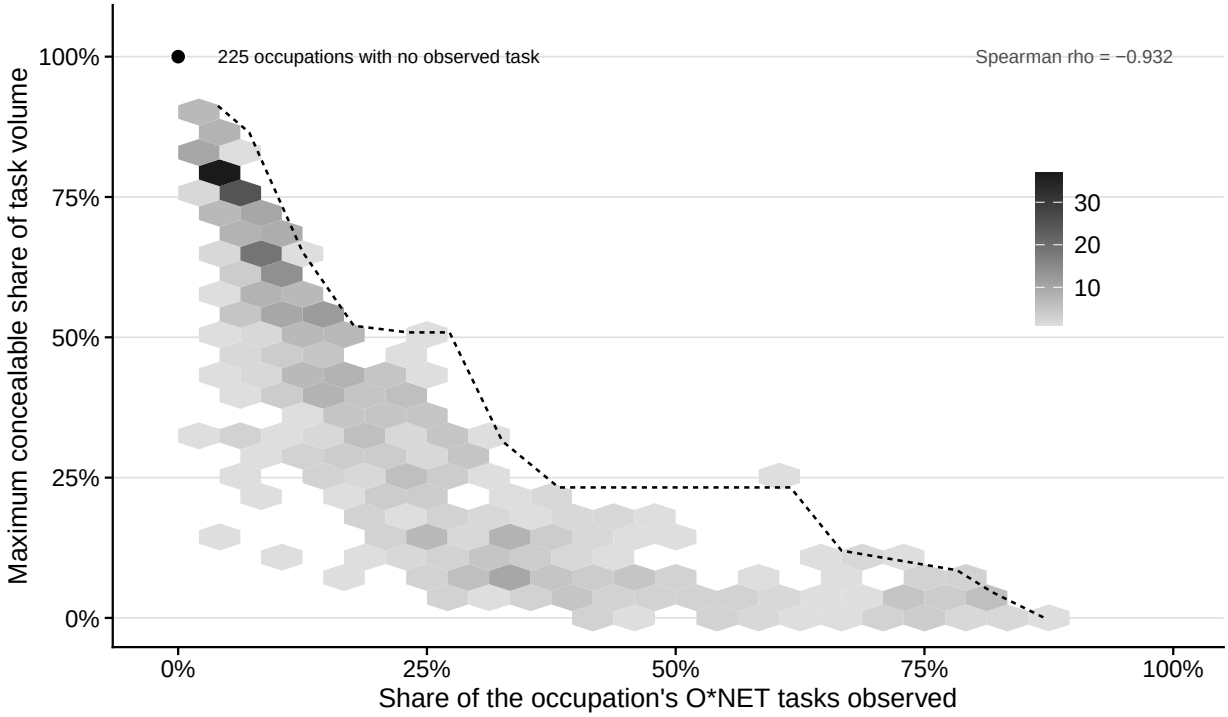


Figure 2: The extensive margin against observed task coverage

Notes: Hexagonal bins over the 550 six-digit SOC occupations with at least one observed task; shading gives the number of occupations per bin. The horizontal axis is the share of an occupation's O*NET tasks appearing in the released data; the vertical axis is h_j, the largest share of its task volume that suppression could be concealing. The dashed line is the empirical upper envelope, computed as a running maximum from the right, and is not a fitted trend. The marked point at the origin represents the 225 occupations (29.0% of 775) with no observed task, for which $h_j = 1$ by construction; they are omitted from the binned surface because a single bin holding 29% of the sample would dominate the shading scale. *Source:* Anthropic Economic Index, release 2025-09-15 (CC-BY 4.0); author's calculations.

median occupation's bound by 0.104 and the mean by 0.138, with a maximum of 0.699. Figure 3 shows the joint identified set for every occupation at once.

The consequence for individual occupations is stark. Of the 91 occupations whose *intensive-margin* bound looked identified at a width below 0.10, 46 do not survive the joint bound. Sheet Metal Workers, Tailors and Multiple Machine Tool Setters all have intensive width exactly zero — every mode present on every observed task — and joint widths of 0.549, 0.332 and 0.322 respectively, because in each case almost nothing of the occupation was observed to begin with. Apparent point identification on the intensive margin is, for these occupations, an artefact of having seen so little.

Figure 2 is worth reading as a frontier rather than as a scatter. Almost nothing sits above the upper envelope, and the envelope is not an empirical regularity but the arithmetic of the censoring rule: given coverage share, there is a maximum concealable volume, and occupations sit at or below it. The rank correlation between coverage share and h_j is -0.932 among the 550 codes with data.

5.4 Identification is a property of the data, not of the work

If bound width tracked something occupational — the character of the work, its routineness, its susceptibility to delegation — then occupational controls should explain variation beyond raw

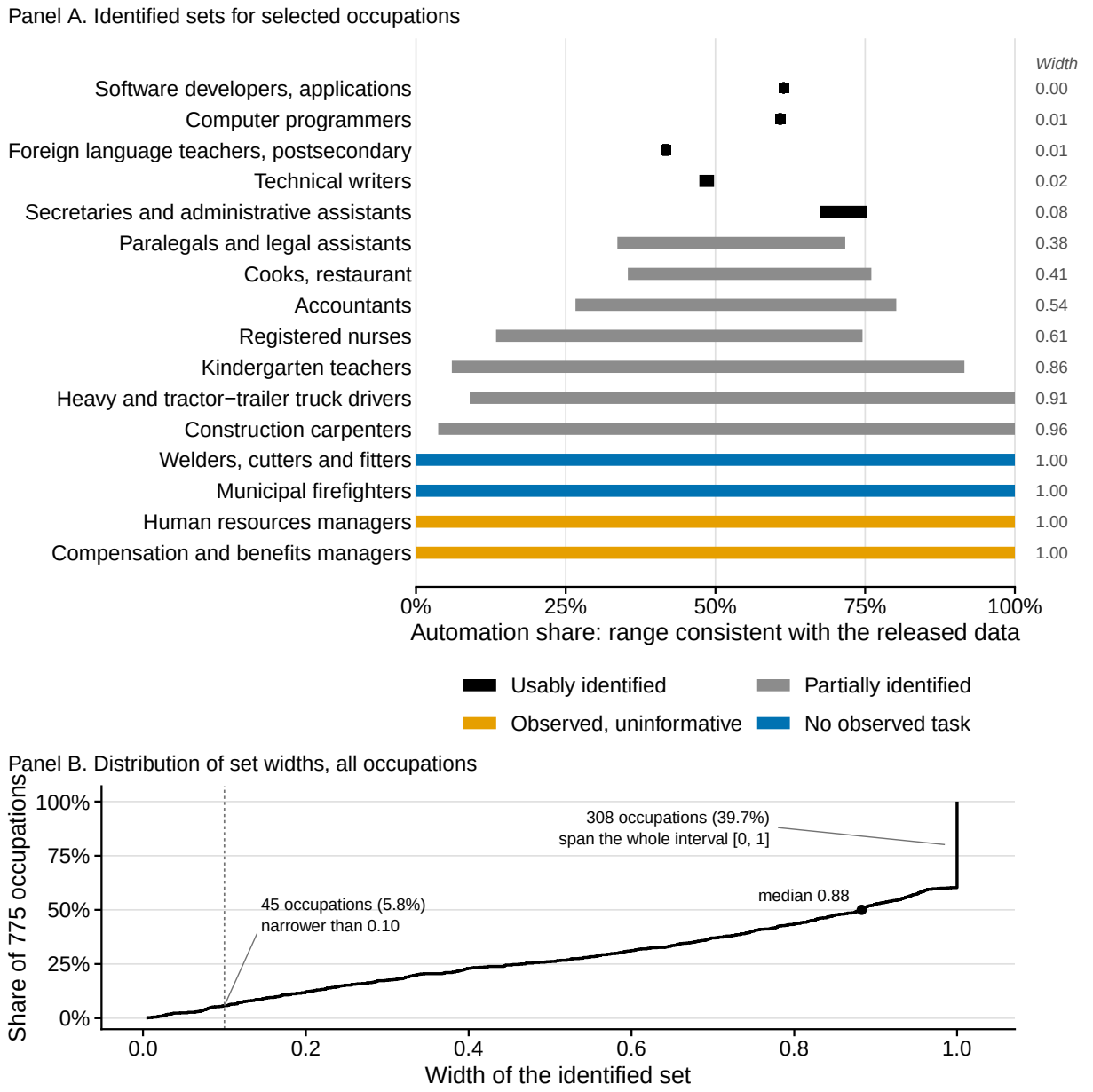


Figure 3: Identified sets for the automation share, all 775 six-digit SOC occupations

Notes: Each vertical segment is one occupation's identified set for the automation share under joint censoring of both margins, ordered left to right by the width of that set. **These are identification bounds, not confidence intervals.** The endpoints are known given the 15-conversation suppression threshold, so the width of a segment reflects what the released data cannot determine rather than sampling variability; no sampling error is depicted. An occupation is counted as usably identified if its set is narrower than 0.10 and it has at least five observed tasks, which holds for 42 of 775; the dashed vertical line marks the last occupation with a set narrower than 0.10. The 308 occupations at the right have sets spanning the entire unit interval and divide into 83 with observed usage that is nonetheless uninformative and 225 with no observed task at all; counts appear above each band. Sixteen of the 22 SOC major groups contain no usably identified occupation.

Source: Anthropic Economic Index, release 2025-09-15 (CC-BY 4.0); author's calculations.

Table 4: Best-identified occupations, joint bounds, at least five observed tasks.

SOC	Title	Tasks	Obs.	Lower	Upper	Width
15-1132	Software Developers, Applications	15	10	0.612	0.616	0.004
15-1131	Computer Programmers	15	11	0.605	0.611	0.006
25-1124	Foreign Language and Literature Teachers	24	18	0.411	0.422	0.011
15-1133	Software Developers, Systems Software	19	12	0.670	0.682	0.012
25-1123	English Language and Literature Teachers	30	22	0.376	0.393	0.017
25-1042	Biological Science Teachers	26	18	0.379	0.397	0.018
15-1142	Network and Computer Systems Administrators	20	9	0.699	0.718	0.020
25-3099	Tutors	19	8	0.364	0.387	0.023
27-3042	Technical Writers	15	8	0.473	0.498	0.025
25-1193	Recreation and Fitness Studies Teachers	23	18	0.219	0.246	0.027

Table 5: Worst-identified occupations, joint bounds, at least five observed tasks.

SOC	Title	Tasks	Obs.	Lower	Upper	Width
11-3111	Compensation and Benefits Managers	24	6	0.000	1.000	1.000
17-3027	Mechanical Engineering Technicians	36	6	0.024	1.000	0.976
39-9031	Fitness Trainers and Aerobics Instructors	22	5	0.000	0.941	0.941
43-5011	Cargo and Freight Agents	55	6	0.021	0.956	0.935
53-1021	First-Line Supervisors of Helpers/Labourers	47	5	0.029	0.927	0.898
43-3021	Statement Clerks	30	7	0.080	0.969	0.889
19-4091	Environmental Science and Protection Technicians	25	5	0.102	0.977	0.875
29-1122	Occupational Therapists	37	6	0.051	0.925	0.874

data density. They do not. Regressing joint bound width on log classified volume across the 550 codes with data gives $R^2 = 0.760$; adding coverage share raises it to 0.843; adding 21 major-group dummies on top of both adds 0.012.[5] The rank correlation between joint width and classified volume is -0.951.

Twenty-one occupational dummies, which between them exhaust the entire major-group structure of the SOC system, add just over one percentage point of explanatory power once we know how much data an occupation has. Identification here is not a fact about jobs. It is a fact about volume.

5.5 Spurious precision

Table 4 lists the best-identified occupations and Table 5 the worst, both restricted to occupations with at least five observed tasks so that neither list is populated by artefacts of thin coverage.

The headline count follows. Forty-five occupations have a joint bound narrower than 0.10, of which 42 — 5.4% of the SOC universe — also rest on at least five observed tasks. Sixteen of the 22 major groups contain none at all, and 28 of the 42, exactly two-thirds, are Computer and Mathematical or Educational Instruction. At the other end, 338 of 775 occupations have an

[5]Specification, which matters and is easy to get wrong: the sample is the 550 SOC codes with at least one observed task, and the volume measure is equal-split observed *classified* conversation volume under $\log(1 + v)$. On all 775 codes, or using total observed volume including censored mass, the first-stage R^2 is 0.613 rather than 0.760. We state this because the paper's own argument is that undocumented data handling produces numbers that cannot be reproduced.

Table 6: Extensive margin by SOC major group, ordered by median concealable volume share h_j. All 775 codes.

Major	Group	Occs	Med. cov.	Med. h_j	Med. width
45	Farming, Fishing, and Forestry	13	0.000	1.000	1.000
47	Construction and Extraction	57	0.000	1.000	1.000
49	Installation, Maint., and Repair	51	0.000	1.000	1.000
51	Production	103	0.000	1.000	1.000
53	Transportation and Material Moving	48	0.000	1.000	1.000
37	Building and Grounds Maintenance	8	0.036	0.787	0.904
33	Protective Service	21	0.039	0.783	1.000
31	Healthcare Support	17	0.059	0.697	0.939
35	Food Preparation and Serving	16	0.087	0.644	0.879
39	Personal Care and Service	30	0.081	0.615	0.928
29	Healthcare Practitioners & Tech.	59	0.130	0.509	0.808
17	Architecture and Engineering	34	0.136	0.471	0.740
43	Office and Administrative Support	52	0.154	0.415	0.748
27	Arts, Design, Entertainment, Media	36	0.188	0.352	0.598
11	Management	34	0.176	0.341	0.651
13	Business and Financial Operations	32	0.231	0.330	0.722
21	Community and Social Service	14	0.226	0.303	0.646
41	Sales and Related	21	0.250	0.243	0.507
23	Legal	8	0.221	0.174	0.351
19	Life, Physical, and Social Science	43	0.258	0.145	0.380
25	Educational Instruction & Library	60	0.691	0.055	0.148
15	Computer and Mathematical	18	0.530	0.022	0.083

identified set spanning $[0.05, 0.95]$, i.e. almost the entire unit interval.

Table 6 gives the extensive-margin picture by major group. In five groups — Farming, Construction, Installation and Repair, Production, and Transportation — the median occupation has *zero* observed tasks. Those five groups contain 272 occupations, 35.1% of the SOC universe, for which the released data supports no statement about automation versus augmentation whatsoever.

6 The near-cancellation: when agreement is not validation

The global automation share published for this release is 51.0698%. That figure is computed over all classified conversations, whether or not they were mapped to an O*NET task, so the comparable object is a reconstruction that retains every row of the `onet_task` by `collaboration` table, including the pseudo-task rows carrying conversations that mapped to no task at all. Summing automation-mode volume over all surviving cells of that table and dividing by surviving classified volume gives 51.6196%.

The gap of $+0.550$ percentage points is the suppression bias, and it decomposes exactly. Of 927,255 classified conversations, 868,120 survive inside cells at an automation share of 51.6196%, which leaves 59,135 suppressed at an automation share of 42.998%. Suppression removed volume markedly less automation-heavy than the volume it left behind, so any reconstruction from surviving cells overstates automation.[6]

[6]The direction is confirmed by decomposition, but the mechanism is narrower than the aggregate suggests, and the

Now consider the reconstruction a researcher would actually want. An occupational measure must exclude the pseudo-tasks, because conversations that map to no O*NET task cannot be attributed to any occupation. That reconstruction gives 51.1477%, i.e. a movement of -0.472 percentage points, and the movement is large because unmapped conversations are markedly more automation-heavy than mapped ones: the two pseudo-task rows together carry 44,928 classified conversations at an automation share of 60.3%.[7]

The two movements are of opposite sign and of nearly equal magnitude; their net is $+0.078$ percentage points. A researcher who correctly excludes the pseudo-tasks, computes an occupational automation share, and then checks it against Anthropic's published global figure will find agreement to within eight hundredths of a percentage point.

This is the uncomfortable part, and it is worth stating precisely, because the natural reading of it is the wrong one. The researcher has not made two mistakes that happened to cancel. The researcher has done the right thing — excluded a non-task from a task universe — and has then compared the result against a quantity computed over a different population. The two figures were never the same object, their agreement is the coincidental cancellation of a suppression bias against a denominator change, and the cancellation survives only in the aggregate. At occupation level the two effects load differently and do not offset, so a pipeline validated this way carries an uncorrected suppression bias into every occupation-level estimate it produces.

Agreement with a published aggregate is therefore not evidence that a reconstruction is correct. In this instance it is close to the opposite, and it may be part of why the censoring documented here has gone unremarked, i.e. the most obvious sanity check available to a downstream user passes.

7 What this means for downstream users

We frame this section as an account of what the released data can support rather than as a claim that published work is wrong. The distinction matters, because for the most part the published work is careful and the problem lies in a property of the data that its documentation does not make salient.

7.1 Occupation-level wage regressions

Specifications that enter automative and augmentative exposure as separate regressors — the approach taken by Chanoi and Bangert-Drowns (2025) — are the most directly affected, because they require the split to be identified at exactly the occupations where our bounds are widest.

natural structural story is wrong. Suppression is not biting the augmentation partition as such: per-mode suppression rates are 6.91% for task iteration, 6.87% for feedback loop, 6.12% for learning and 4.97% for directive, a band that crosses the automation–augmentation line freely and in which the two sides interleave. Validation is the exception at 15.94%, more than double any other mode, and it accounts for 87% of the net excess augmentation-side suppression; setting its rate to the mean of the remaining four reduces the gap from $+0.550$ to $+0.301$ percentage points. Validation is the rarest mode by volume, at 4.7% of the total spread across the same 2,616 tasks, so its cells are the thinnest and clear the 15-conversation floor least often. The mechanism is cell thinness operating through one mode's scarcity rather than a structural property of the partition. This has a substantive corollary: validation is the mode in which a human checks model output, so suppression selectively erases the record of humans validating AI work at roughly two and a half times the rate it erases anything else.

[7] Derived as the difference between the with- and without-pseudo-task reconstructions. The saved artifacts do not separate `none` from `not_classified`, so we report the pair rather than attributing the mass to either; the six intersection rows involved would need to be split at source to say more.

Where two exposure coefficients are estimated to be near-equal and opposite, the relevant question is whether the identified set for the underlying split admits both signs, and for the majority of occupations it does. Propagating our bounds through such a specification is the single strongest robustness exercise available here, and we regard it as the natural sequel rather than as something this paper establishes.[8]

7.2 Regional exposure indices

Regional indices employment-weight national occupation scores to local geography. The Tufts American AI Jobs Risk Index (2026) uses the AEI alongside capability-based measures and works explicitly in automation terms; the MIT Center for Transportation and Logistics AI Labor Exposure Map (2026) draws current-capability estimates from the AEI and weights them with BLS employment at state and metropolitan level. Neither, however, publishes any measure of uncertainty on its rankings.

The consequence is a specific and, we think, serious distributional problem. Because these indices apply national occupation scores to local employment, a region's score is a weighted average of occupational scores whose weights are local but whose values are national. Regions whose employment concentrates in the five major groups where the median occupation has no observed tasks — farming, construction, installation and repair, production, transportation — receive scores assembled almost entirely from occupations for which the underlying parameter is unidentified. Those are disproportionately rural and lower-wage regions, and they are the intended targets of workforce-transition funding. We do not claim that the direction of any published regional ranking is wrong; we claim that fine distinctions between similarly-ranked regions are not identified, and that the unidentified portion is not randomly distributed across the country.

7.3 Recode-to-manual

Demirer et al. (2026) recode fully-filtered tasks to manual and note that this limits statistical power. In the framework here, that recode is a point in the identified set, specifically the lower endpoint $\underline{\theta}$, chosen by assumption. The quantity that matters is how far that endpoint sits from the other one, and Table 6 supplies it: for the median occupation the distance is 0.883, and for the 225 occupations with no observed task it is the whole interval.

7.4 Skill-level and within-occupation designs

A natural response to the occupation-level results is to move the analysis down a level, to the skills advertised in vacancies, and to assign each skill the exposure of the tasks it serves. That relocates the problem rather than escaping it. A skill's measure is an aggregate of task-level measures taken through a crosswalk, so it inherits the identified sets of the tasks it is linked to, and aggregation narrows those sets only to the extent that censoring is imperfectly correlated across the linked tasks. The occupation-level results indicate how little that buys: pooling tasks into occupations narrows the median intensive-margin bound from 0.5411 to 0.4177, by about a fifth. A skill linked to a single task carries that task's identified set unchanged, and nothing in the aggregation narrows it.

[8] We do not report coefficient magnitudes from that literature, because we have not verified them against the published tables.

Comparisons of differently exposed skills within an occupation are attractive for a separate reason, which is that occupation-by-time effects absorb the occupation-level demand shocks that confound comparisons across occupations. They face a different difficulty, and it follows directly from Section 5.4. Measurement precision in these data is close to a deterministic function of usage volume, and usage volume is itself correlated with exposure: text-mediated work is both more exposed to generative AI and more heavily represented in data on its use. Within an occupation, then, the highly exposed tasks and skills are the precisely measured ones, while the less exposed comparison group is the one most often suppressed — and suppressed cells read as zeros understate precisely the usage against which the exposed group is being compared.

The measurement error this produces is non-classical. It is concentrated in one arm of the comparison and correlated with the regressor of interest, so the standard attenuation result does not apply, and the sign of the resulting bias is not determined without further assumptions about the relationship between outcomes and true exposure. We think the correlation between precision and exposure should be treated as a first-order threat to within-occupation designs rather than as noise to be averaged away.

7.5 Rubric-scored exposure is not a free substitute

The obvious response to this paper is to abandon observed usage for rubric-based scoring, in which a model or a panel rates each O*NET task directly, as Eloundou et al. (2024) do for exposure. The substitution has real advantages: a rubric scores every task, including those that suppression removes, and it returns point values rather than intervals. It does not escape the identification problem, however. It changes the form the problem takes.

Observed usage data fail through suppression and, bounded correctly, say so, in the form of wide identified sets. Judged data return a value for every task, but those values are identified only up to the choice of rater, and there is direct evidence that the choice matters. Yin, Vu and Persico (2026) apply one rubric with three frontier models to identical tasks and find mean exposure varying 3.6-fold, with pairwise agreement as low as 57%. Downstream, individual-level difference-in-differences estimates agree in sign across raters but vary 2.4-fold in magnitude, while county-level estimates move from a significant negative to an insignificant positive depending on which model scored the tasks.

These are two routes to an uninformative measure, and they differ in what they disclose. A wide identified set announces its own uninformativeness. A point estimate does not: it carries no record of how far a different rater would have moved it, and a set of rubric scores that barely discriminate between tasks looks exactly as authoritative as one that discriminates well.

The natural response is to validate judged scores against observed ones, and Yin, Vu and Persico do exactly that, reporting rank correlations with occupation-level AEI usage of roughly 0.3 to 0.4 across raters. The results here complicate the reading of that comparison. It treats observed usage as a point value, which for a large share of occupations it is not, and a benchmark in which suppressed cells appear as zeros can depress any rank correlation computed against it. Modest agreement is therefore consistent with rater instability, with censoring in the benchmark, or with both, and a comparison against point values cannot separate them.

Separating them requires validating against identified sets, and specifically against the sharp ones. A rubric score will fall inside a wide identified set almost regardless of its accuracy, so consistency with the observed data is informative only where the set is narrow. The 238 point-

identified tasks in this release are the natural validation sample, stratified by identification quality so that agreement on the sharp cases is not diluted by vacuous agreement on the wide ones. We leave that exercise to future work.

8 Recommendations

Three changes would restore most of what suppression removes, and none requires relaxing the privacy protection.

Publish suppression flags. The released file cannot currently be distinguished from one in which a cell is genuinely zero. A boolean marking each absent cell as suppressed rather than empty would convert a silent point estimate into an interval that any user could construct, at no disclosure cost.

Publish marginal totals. Cell suppression protects individual cells; the row and column totals it protects are typically publishable. Task-level totals would bound the censored mass far more tightly than the blanket $C = 14$ used here.

Adopt partial suppression. Fischetti and Salazar-González (2003) show how to publish intervals in place of suppressed cells while preserving the disclosure guarantee. Applied here, it would make the exercise in this paper unnecessary, which is the outcome we would prefer.

For users of the current release, the recommendation is narrower: report the number of observed tasks alongside any occupation-level share, treat occupations below roughly five observed tasks as unidentified, and do not interpret differences between occupations whose identified sets overlap.

9 Limitations

The bounds are worst-case and make no assumption about how suppressed volume distributes across modes. A researcher willing to assume that suppressed cells resemble observed cells within the same task would obtain far narrower intervals. We regard the worst-case bound as the right object for a paper whose point is that the assumption is currently made implicitly and without acknowledgement, but it is an upper bound on ignorance rather than a best estimate.

The censoring cap is treated as binding on the conversation criterion alone, which as noted in Section 3.4 makes the bounds conservative.

The results rest on a single release, a single week of sampling, the Claude.ai source, and the global aggregate. Whether the pattern replicates across the January, March and June 2026 releases is the obvious question and is not answered here.

The occupation-level bounds are unweighted across occupations. Employment weighting with OEWS would convert an occupational result into a labour-market one, and in particular would allow the statement that some share of *employment* sits in unidentified occupations, which is the quantity a policy reader wants and which we cannot currently supply.

Finally, the equal-split convention for allocating task volume across occupations is inherited from the index's own construction. It is a strong assumption, and the bounds would change under any alternative, though we see no reason to expect the qualitative pattern to change with it.

10 Conclusion

The automation–augmentation split in the Anthropic Economic Index is the most direct public evidence we have about how AI is actually used rather than about how it might be used, and that

is precisely why the measure deserves the scrutiny applied here. Privacy suppression removes 6.38% of classified conversation volume, which sounds small and is not: because suppression concentrates in thin cells and most tasks are thin, it removes identification from the great majority of the occupational distribution. Forty-two of 775 occupations retain a usably narrow identified set, sixteen of 22 major groups contain none, and 338 occupations have identified sets covering almost the whole unit interval.

The gradient is the part that should worry a policy reader. Identification survives where work passes through text and therefore through a chat interface, and it fails where work is manual. Those are not symmetric failures. The occupations where the automation-versus-augmentation question is politically and economically live — where displacement would be most disruptive and least easily absorbed — are exactly the occupations about which the released data can say nothing. That the index is nonetheless being employment-weighted into county-level rankings intended to direct transition funding is a problem, not because the rankings are necessarily wrong, but because nobody currently reports how much of them is assumption.

References

The Budget Lab at Yale (2026). Labor market AI exposure: what do we know? Yale University.

Brynjolfsson, E., Chandar, B., & Chen, R. (2025). Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence. Stanford Digital Economy Lab working paper.

Brynjolfsson, E., Mitchell, T., & Rock, D. (2018). What can machines learn, and what does it mean for occupations and the economy? *AEA Papers and Proceedings*, 108, 43–47.

Chanoi, C., & Bangert-Drowns, C. (2025). AI exposure by U.S. occupations and work tasks and the effect on wages. Washington Center for Equitable Growth working paper.

Cox, L. H. (1980). Suppression methodology and statistical disclosure control. *Journal of the American Statistical Association*, 75, 377–385.

Cox, L. H. (1995). Network models for complementary cell suppression. *Journal of the American Statistical Association*, 90, 1453–1462.

Demirer, M., Horton, J., Immorlica, N., Lucier, B., & Shahidi, A. (2026). Chaining tasks, redefining work: a theory of AI automation. *NBER Working Paper* 34859.

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: labor market impact potential of LLMs. *Science*, 384(6702), 1306–1308.

Felten, E., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence. *Strategic Management Journal*, 42(12), 2195–2217.

Fischetti, M., & Salazar-González, J. J. (2003). Partial cell suppression: a new methodology for statistical disclosure control. *Statistics and Computing*, 13, 13–21.

Frey, C. B., & Osborne, M. A. (2017). The future of employment: how susceptible are jobs to computerisation? *Technological Forecasting and Social Change*, 114, 254–280.

Handa, K., Tamkin, A., McCain, M., et al. (2025). Which economic tasks are performed with AI? Evidence from millions of Claude conversations. arXiv:2503.04761.

Horowitz, J. L., & Manski, C. F. (1995). Identification and robustness with contaminated and corrupted data. *Econometrica*, 63(2), 281–302.

Imbens, G. W., & Manski, C. F. (2004). Confidence intervals for partially identified parameters. *Econometrica*, 72(6), 1845–1857.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer.

Manski, C. F., & Tamer, E. (2002). Inference on regressions with interval data on a regressor or outcome. *Econometrica*, 70(2), 519–546.

Massenkoff, M., Lyubich, E., McCrory, P., Appel, R., & Heller, R. (2026). Anthropic Economic Index report: learning curves. Anthropic, March 2026.

MIT Center for Transportation and Logistics (2026). AI Labor Exposure Map.

Stoye, J. (2009). More on confidence intervals for partially identified parameters. *Econometrica*, 77(4), 1299–1315.

Tamkin, A., McCain, M., Handa, K., et al. (2024). Clio: privacy-preserving insights into real-world AI use. arXiv:2412.13678.

Tufts University Digital Planet (2026). American AI Jobs Risk Index. The Fletcher School.

Webb, M. (2020). The impact of artificial intelligence on the labor market. Working paper.

Yin, M., & Ogut, B. (2026). Who uses AI? Platform selection and the measurement of occupational AI exposure. arXiv:2605.21743.

Yin, M., Vu, H., & Persico, C. (2026). How (un)stable are LLM occupational exposure scores? Evidence from multi-model replication. NBER Working Paper 35110.

Appendix A: reproduction

All results derive from a single script, `03-suppression-bounds.R`, run under R 4.6.1 with `aieconindex` 0.2.0. Raw files are fetched at runtime from the Hugging Face repository; no manual steps intervene. The script carries assertions at each failure-prone join, a censoring sensitivity sweep, and a drift table comparing fifteen metrics against a reference run. At the run reported here all fifteen matched exactly and the task-to-O*NET join rate was 100%.

Four errors were found and fixed during development and are recorded in the script header: a selection-on-outcome error in the bounds construction, a major-group collapse in the mode composition, an unfiltered `none` pseudo-task, and a denominator mislabel. The regression and rank correlations reported in Section 5.4 are computed from `occ_extensive_joint.rds` and are not currently emitted by the script; extending it to do so, and adding them to the drift table, is the first item of outstanding work.